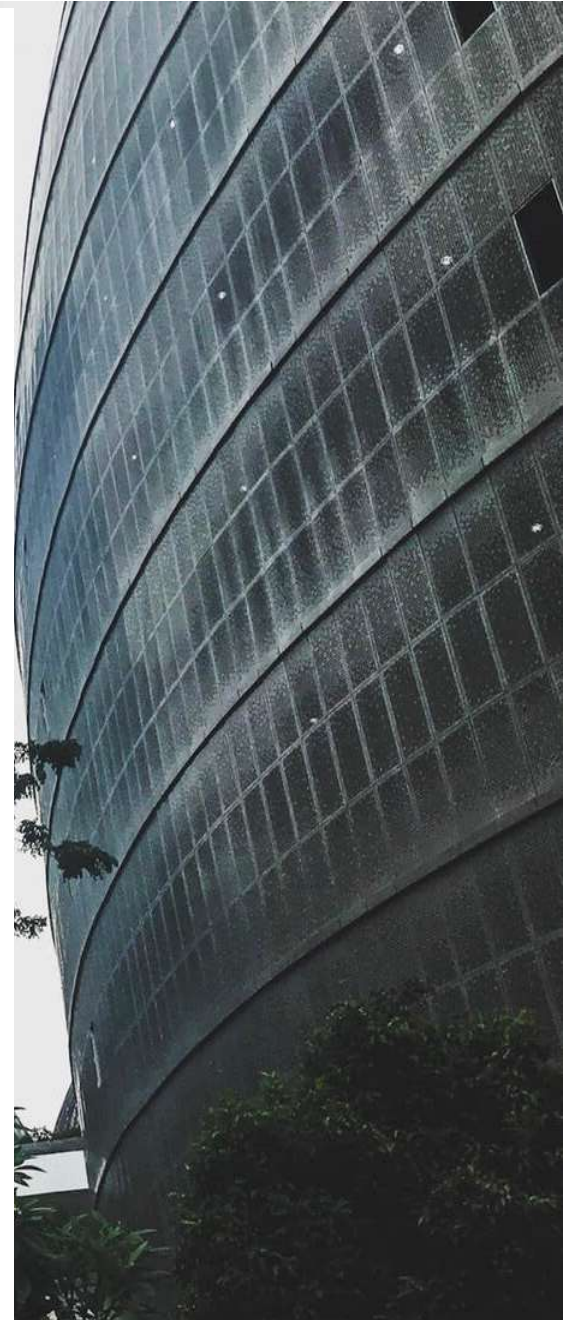


COURSE MATERIAL

**IF670 – Mobile Cross Platform Programming
S1 INFORMATICS GRADUATE PROGRAM
FACULTY OF ENGINEERING AND INFORMATICS**

**INFORMATICS STUDY PROGRAM
FACULTY OF ENGINEERING AND INFORMATION
UNIVERSITAS MULTIMEDIA NUSANTARA**

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MODUL 5

(React Native Navigation)

THEME DESCRIPTION

Students will learn about React Native Navigation

WEEKLY LEARNING GOAL (SUB-GOAL LEARNING)

Students will be able to understand and implement basic routing and dynamic routing in cross-platform application development.

SOFTWARE / HARDWARE REQUIREMENTS

1. Laptop / PC
2. IDE (Android Studio/XCode)
3. Code Editor (Visual Studio Code/Sublime/etc)

PRACTICUM STEPS

1. Create a new project based on your Week 4 project by copying its source code, and then create a new repository for the new project for this week.
2. Install @react-navigation/native on your project with this command:
npm install @react-navigation/native
3. Install dependencies for expo with this command:
npx expo install react-native-screens react-native-safe-area-context
4. Install 'native-stack' library with this command:
npx expo install @react-navigation/native-stack
5. Create a new file named **home.tsx** and write the following code:

```
import { Link } from "expo-router";
import { Button, Text, View } from "react-native";

export default function home() {
  return (
    <View style={{flex:1, alignItems:"center",justifyContent:"center"}}>
      <Text>Navigation List</Text>
      <Link href="/email" push asChild>
        <Button
          title="Go to Email Screen"
        />
      </Link>
    </View>
  );
}
```

6. Create a new file named **email.tsx** and write the following code:

```
import { Link } from "expo-router";
import { Button, Text, View } from "react-native";
import styles from "../AppStyles";

export default function email() {
  return (
    <View style={styles.container}>
      <Text>Email List Page</Text>
      <Link href="/home" push asChild>
        <Button
          title="Go to Home Screen"
        />
      </Link>
    </View>
  );
}
```

7. Move the user list from **index.tsx** into a new file named **userList.tsx**

```
import { ScrollView, View } from "react-native";
import { Avatar, Card, Text } from "react-native-paper";
import styles from "../AppStyles";
import userData from "../data.json";

export default function userList() {
  return (
    <ScrollView contentContainerStyle={styles.container}>
      {userData.map((users, index) => (
        <Card key={index} style={styles.card}>
          <Card.Content style={styles.cardContent}>
            <Avatar.Image
              size={70}
              source={{ uri: users.photo_url }}
            />
            <View style={styles.textContainer}>
              <Text variant="titleMedium">
                {users.name}
              </Text>
              <Text variant="bodyMedium">
                {users.email}
              </Text>
            </View>
          </Card.Content>
        </Card>
      ))}
    </ScrollView>
  );
};
```

8. Update your **index.tsx** file based on the code provided below. Were you successfully redirected to the home page?

```
import { Link, Stack } from 'expo-router';
import { View } from 'react-native';

export default function App() {
  return (
    <>
      <Stack.Screen options={{title:"Welcome"}}/>
      <View style={{flex:1, alignItems:"center",justifyContent:"center"}}>
        <Link href="/home">GO TO NAVIGATION LIST</Link>
      </View>
    </>
  );
}
```

9. Create a new file named **profile.tsx** and write the following code:

```
import { Link, useLocalSearchParams } from "expo-router";
import { Button, Text, View } from "react-native";

export default function home() {
  const { userName } = useLocalSearchParams<{ userName: string }>();
  return (
    <View style={{flex:1, alignItems:"center",justifyContent:"center"}}>
      <Text>{userName}&apos;s Profile</Text>
      <Link href="/home" push asChild>
        <Button
          title="Go to Home Screen"
        />
      </Link>
    </View>
  );
}
```

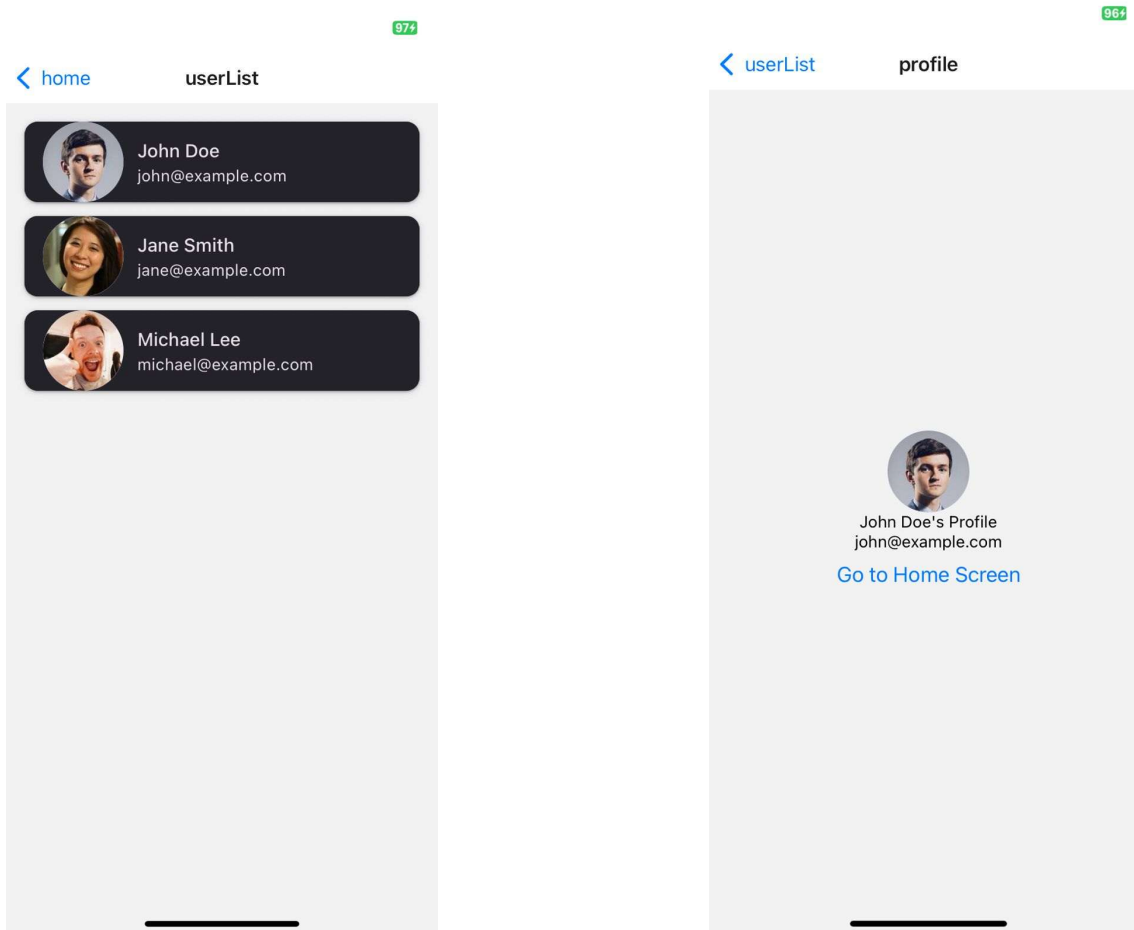
10. Modify your `userList.tsx` file so that it supports dynamic routing for each user.

```
import { Link } from "expo-router";
import { ScrollView, TouchableOpacity, View } from "react-native";
import { Avatar, Card, Text } from "react-native-paper";
import styles from "../AppStyles";
import userData from "../data.json";

export default function userList() {
  return (
    <ScrollView contentContainerStyle={styles.container}>
      {userData.map((users, index) => (
        <Card key={index} style={styles.card}>
          <Link href={{
            pathname: "/profile",
            params: { userName: users.name }
          }} push asChild>
            <TouchableOpacity
              style={styles.card}
            >
              <Card.Content style={styles.cardContent}>
                <Avatar.Image
                  size={70}
                  source={{ uri: users.photo_url }}
                />
                <View style={styles.textContainer}>
                  <Text variant="titleMedium">
                    {users.name}
                  </Text>
                  <Text variant="bodyMedium">
                    {users.email}
                  </Text>
                </View>
              </Card.Content>
            </TouchableOpacity>
          </Link>
        </Card>
      ))}
    </ScrollView>
  );
};
```

ASSIGNMENT

Add "Go to User List Page" in `home.tsx`. Then update `userList.tsx` and `profile.tsx` to display the complete user data, matching the example result shown below.



For the submission:

1. Create a Git repository with the name Week5_[NIM]_[Name].
2. Push your project files to the repository.
3. Copy the repository link and paste it into Week5_[NIM]_[Name].txt
4. Submit to elearning in .txt or .zip format

REFERENSI

1. <https://docs.expo.dev/tutorial/add-navigation/>